

Amulya Murahari

USA — 682-376-0562 — murahariamulya@gmail.com — [GitHub](#) — [LinkedIn](#) — [Portfolio](#)

EXPERIENCE

Software Engineer

Apr 2025 – Present

Comcast - Cppgemini

Philadelphia, USA

- Leading the design and development of an endurance regression framework, reproducing customer-reported field escapes as automated pre-release validation to improve release quality
- Developed AI-assisted visual validation workflows using DIOCR and image recognition improving automation resilience across remote-controlled streaming devices
- Built annotated datasets and fine-tuned YOLO models for AI-driven recognition of application states and UI elements across TV applications

Software Engineer

Oct 2024 – Apr 2025

Changing The Present

San Jose, USA

- Built Next.js and TypeScript product workflows to accelerate stakeholder feedback and validate user-facing features
- Engineered a high-performance search and discovery experience using Elasticsearch with filtered search and type-ahead suggestions, reducing search latency and boosting conversion rates by 25%
- Designed and maintained Jenkins CI/CD pipelines using Docker containers and Kubernetes, reducing average deployment time from 20–30 minutes to under 7 minutes

Software Engineer

Jan 2024 – May 2024

Renavest

Boston, USA

- Developed an AI-driven personalized learning module using OpenAI API, FastAPI, and React, adapting content based on user behavior and skill level—resulting in a 20% boost in user engagement and retention
- Designed scalable REST APIs using Golang to capture user interactions and behavioral signals, enabling real-time recommendations for the AI learning engine
- Integrated Apache Kafka with FastAPI to stream user events, improving personalization accuracy and reducing session lag for high-volume learners

Associate Software Engineer

Dec 2020 – Jun 2022

Accenture

Bangalore, India

- Developed RESTful microservices using Java and Spring Boot for data validation and integration with HR systems, reducing manual data entry time by 15%
- Built a serverless file processing pipeline using AWS S3, Lambda, and Step Functions to extract metadata, convert formats, and support scalable document uploads
- Optimized SQL queries using Hibernate and Redis caching, improving RDS query response times by 50%

EDUCATION

Masters in Computer Software Engineering, Northeastern University

Aug 2024

Coursework: Program Structures and Algorithms, Object-Oriented Design, Prompt Engineering and Gen AI, Web Design

Bachelors in Computer Science, Visvesvaraya Technological University (VTU)

Aug 2020

Coursework: Algorithm Design, Machine Learning, Operating Systems, Cloud Computing

SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL, C/C++, C#
Frontend	React, Next.js, Angular, React Native, HTML5, CSS3
Backend	Node.js, Express.js, Spring Boot, FastAPI, REST APIs
AI/ML	LLMs, Prompt Engineering, RAG, AI Agents, Google Vertex AI, YOLO, OpenAI API, DIOCR
Database	PostgreSQL, MySQL, MongoDB, Redis, SQLite, DynamoDB, Oracle DB
Developer Tools	Git, GitHub, Postman, Cypress, Jest, Maven, Linux, Unix, Webpack, Vite
Cloud Services	Azure (DevOps), AWS (Lambda, ECS, S3, DynamoDB), Google Cloud Platform

PROJECT

Aqua Paths

Apr 2024

- Utilized Sea Routes APIs to develop an open platform optimizing sea routes, achieving a 40% reduction in CO2 emissions by factoring in weight and cargo. Implemented Marine Life insights tool